ESE5180: Lab 0 Zephyr

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GitHub Repository URL: https://github.com/master869/lab0-zephyr-master869.git

1. Hello (Vanilla) Zephyr

(1.1) Create a video showing blinky on all 3x MCU boards.

- In the video, show your terminal history of compiling and flashing for all 3x MCU boards.
- In the video, show each MCU board flashing its LED while running the blinky code.
- Name the video file: **f26_lab0_1.1_pennkey**
- Submit the video to this Google Form.

I have already submitted the video to the Google Form, and you can also watch it through this link

2. Hello (Nordic) Zephyr

(2.1) Commit your Zephyr application to your GitHub repository

Done

(2.2) Create a video showing the change in blinky behavior on the nRF7002DK

- In the video, show your terminal history within VS Code
- In the video, show the nRF7002DK board blinking at its original rate, then change the blinking time to 2 seconds, flash, and show the execution.
- Name the video file: **f26_lab0_2.1_pennkey**
- Submit the video to this Google Form.

I have already submitted the video to the Google Form, and you can also watch it through this link

3. Building with West

(3.1) Build and Flash using only west commands and not the GUI. Show the terminal prints by embedding a screenshot in your README.md.

```
master@Mac blinky % west build
[0/8] Performing build step for 'blinky'
[0/18] Performing build step for 'tfm'
ninja: no work to do.
[2/10] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[10/10] Linking C executable zephyr/zephyr.elf
Memory region         Used Size  Region Size  %age Used
           FLASH:       22576 B     188080 B     12.00%
             RAM:        4600 B       192 KB      2.34%
        IDT_LIST:           0 B        32 KB      0.00%
Generating files from /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/build/blinky/zephyr/zephyr.elf for board: nrf7002dk/nrf5340/cpuapp/ns
image.py: sign the payload
image.py: sign the payload
[1/8] Performing build step for 'mcuboot'
ninja: no work to do.
[8/8] Completed 'mcuboot'
master@Mac blinky % west flash
-- west flash: rebuilding
[0/8] Performing build step for 'blinky'
[0/16] Performing build step for 'tfm'
ninja: no work to do.
[2/3] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[3/3] Completed 'tfm'
[1/8] Performing build step for 'mcuboot'
ninja: no work to do.
[7/7] Completed 'mcuboot'
-- west flash: using runner nrfutil
Using board 001050785123
-- runners.nrfutil: Flashing file: /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/build/mcuboot/zephyr/zephyr.hex
-- runners.nrfutil: Connecting to probe
-- runners.nrfutil: Erasing address ranges touched by firmware
-- runners.nrfutil: Programming image
-- runners.nrfutil: Verifying image
-- runners.nrfutil: Board(s) with serial number(s) 1050785123 flashed successfully.
-- west flash: using runner nrfutil
-- runners.nrfutil: reset after flashing requested
-- runners.nrfutil: Flashing file: /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/build/blinky/zephyr/zephyr.signed.hex
-- runners.nrfutil: Connecting to probe
-- runners.nrfutil: Erasing address ranges touched by firmware
-- runners.nrfutil: Programming image
-- runners.nrfutil: Verifying image
-- runners.nrfutil: Reset
-- runners.nrfutil: Board(s) with serial number(s) 1050785123 flashed successfully.
```

4. Kconfig

5. Device Tree (DT)

(5.1) Switch out the blinking to **LED2**. Create an overlay file with your own alias named **LED5180** that links to **LED2** on the nRF7002 DK. Ensure you only call this alias in the main.c program. Commit your code to this GitHub Classroom repository.

Done

(5.2) Poll for a button press and switch the LED state. Commit these application changes to your GitHub repository.

Done

(5.3) Create a new alias for your button and call your alias within your main.c. Commit these application changes to your GitHub repository.

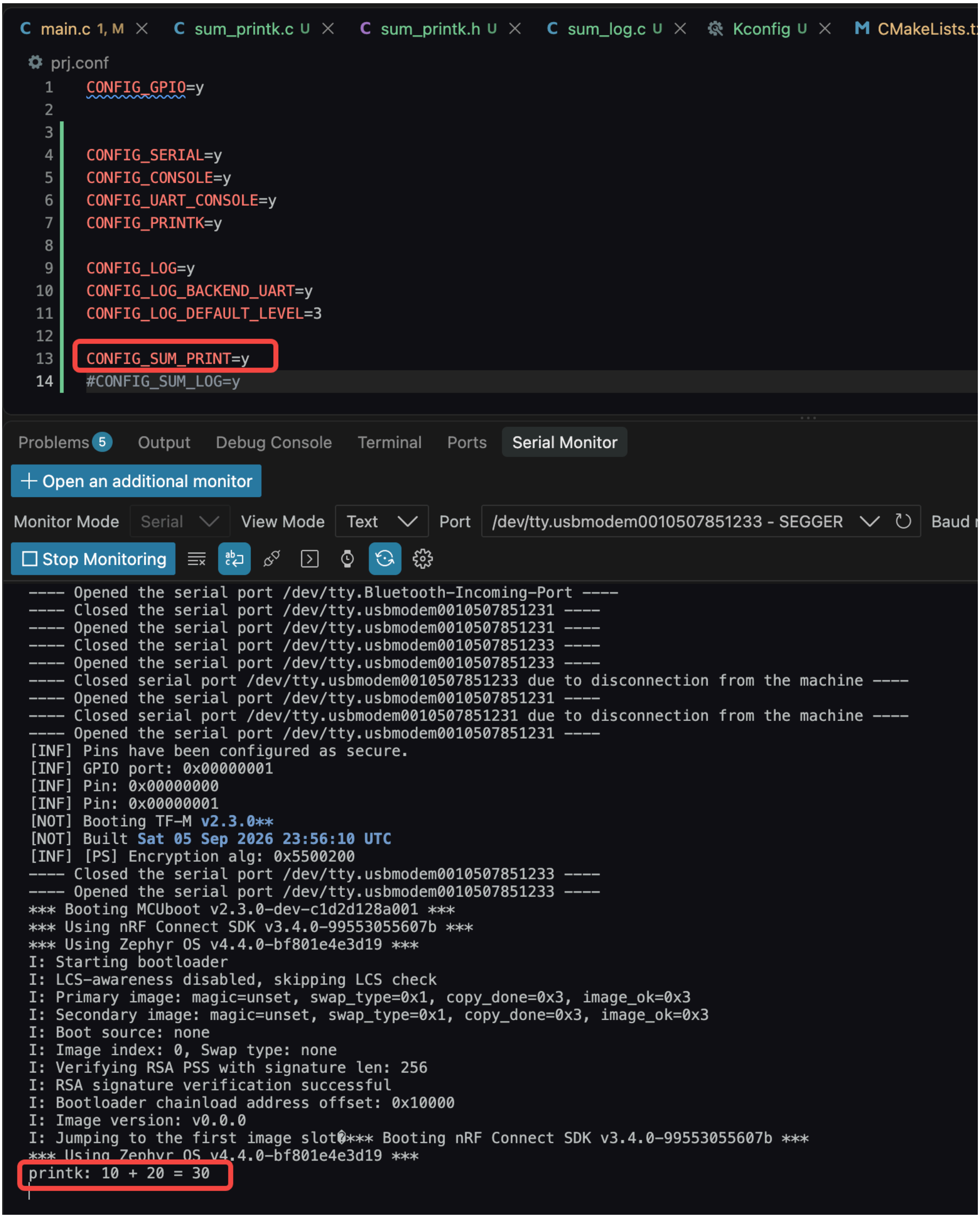
Done

6. Printing vs. Logging

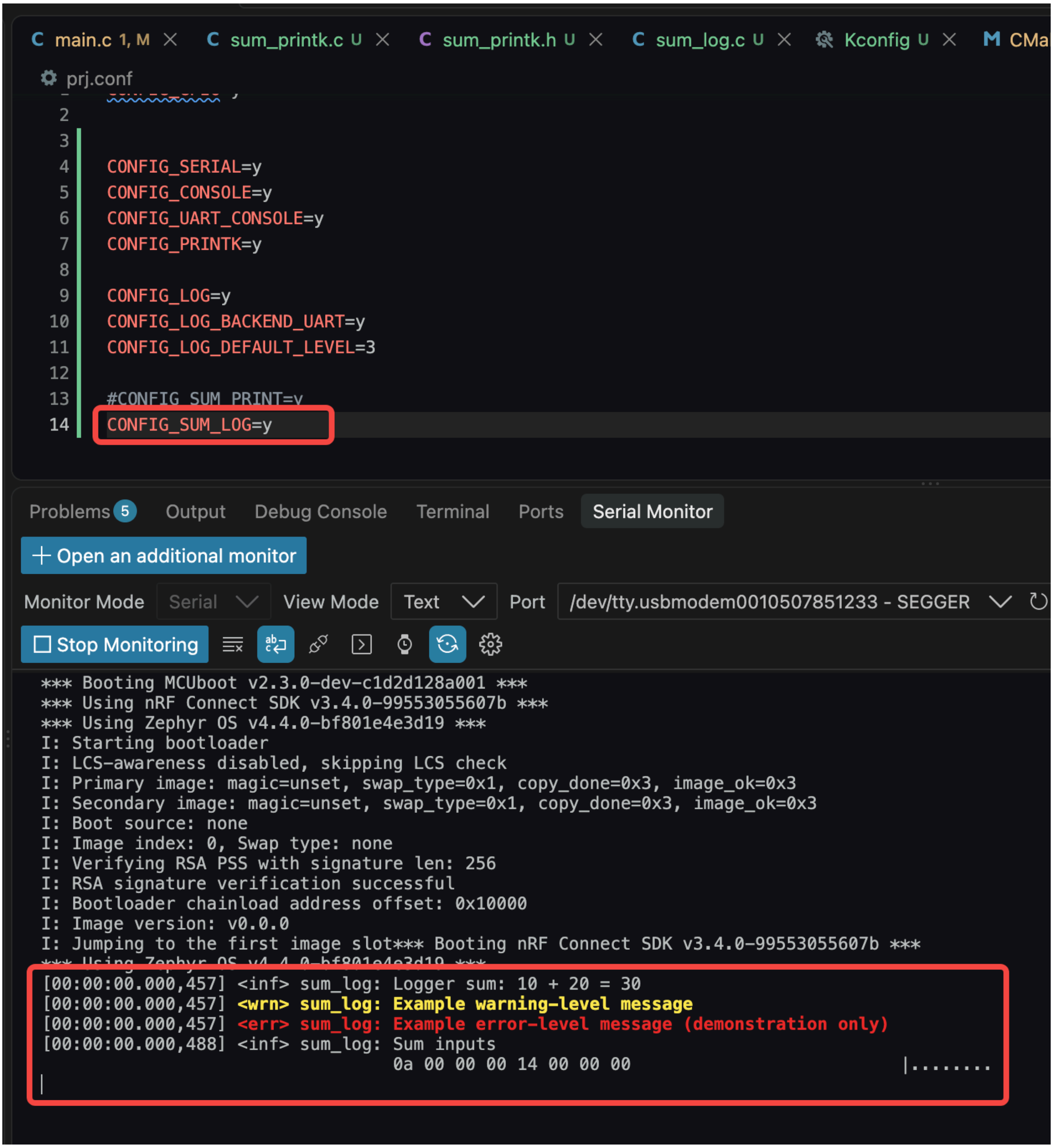
(6.1) Take screenshots of console output for both builds:

- `CONFIG_SUM_PRINT=y` → result printed with `printk()`.
- `CONFIG_SUM_LOG=y` → result printed with the `Logger` (include hexdump).

printk():


main.c 1, M
sum_printk.c U
sum_printk.h U
sum_log.c U
Kconfig U
CMakeLists.t
prj.conf
CONFIG_GPIO=y
CONFIG_SERIAL=y
CONFIG_CONSOLE=y
CONFIG_UART_CONSOLE=y
CONFIG_PRINTK=y
CONFIG_LOG=y
CONFIG_LOG_BACKEND_UART=y
CONFIG_LOG_DEFAULT_LEVEL=3
CONFIG_SUM_PRINT=y
#CONFIG_SUM_LOG=y
Problems 5
Output
Debug Console
Terminal
Ports
Serial Monitor
Open an additional monitor
Monitor Mode
Serial
View Mode
Text
Port
/dev/tty.usbmodem0010507851233 - SEGGER
Stop Monitoring
---- Opened the serial port /dev/tty.Bluetooth-Incoming-Port ----
---- Closed the serial port /dev/tty.usbmodem0010507851231 ----
---- Opened the serial port /dev/tty.usbmodem0010507851231 ----
---- Closed the serial port /dev/tty.usbmodem0010507851233 ----
---- Opened the serial port /dev/tty.usbmodem0010507851233 ----
---- Closed serial port /dev/tty.usbmodem0010507851233 due to disconnection from the machine ----
---- Opened the serial port /dev/tty.usbmodem0010507851233 ----
---- Closed serial port /dev/tty.usbmodem0010507851231 due to disconnection from the machine ----
---- Opened the serial port /dev/tty.usbmodem0010507851231 ----
[INF] Pins have been configured as secure.
[INF] GPIO port: 0x00000001
[INF] Pin: 0x00000000
[INF] Pin: 0x00000001
[NOT] Booting TF-M v2.3.0**
[NOT] Built Sat 05 Sep 2026 23:56:10 UTC
[INF] [PS] Encryption alg: 0x5500200
---- Closed the serial port /dev/tty.usbmodem0010507851233 ----
---- Opened the serial port /dev/tty.usbmodem0010507851233 ----
*** Booting MCUboot v2.3.0-dev-c1d2d128a001 ***
*** Using nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
I: Starting bootloader
I: LCS-awareness disabled, skipping LCS check
I: Primary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Secondary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Boot source: none
I: Image index: 0, Swap type: none
I: Verifying RSA PSS with signature len: 256
I: RSA signature verification successful
I: Bootloader chainload address offset: 0x10000
I: Image version: v0.0.0
I: Jumping to the first image slot*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
printk: 10 + 20 = 30

`Logger:`



(6.2) Make a short video showing hexdump, log, and printk.

- Name the video file: **f26_lab0_6.2_pennkey**
- Submit the video to this Google Form.

I have already submitted the video to the Google Form, and you can also watch it through this link

(6.3) Commit your updated Zephyr application to your GitHub repository

Done

7. Ztest for Unit Testing

(7.1) Implement the test case in "TO DO" to test your function. Commit your updated Zephyr application to your GitHub repository

Done

(7.2) Take screenshots of the testing outputs (laptop) in the terminal.

A: run in test directory (SUM_UNIT_TEST)

```
master@Mac SUM_UNIT_TEST % west build -t run
-- west build: running target run
[0/1] To exit from QEMU enter: 'CTRL+a, x'[QEMU] CPU: cortex-m3
qemu-system-arm: warning: nic stellaris_enet.0 has no peer
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE sum_log_test_suite
===================================================================
START - test_sum_log_basic
I: Logger sum: 10 + 20 = 30
W: Example warning-level message
E: Example error-level message (demonstration only)
I: Sum inputs
I: 0a 00 00 00 14 00 00 00 |........

   Assertion succeeded at CMAKE_SOURCE_DIR/src/sum_test.c:8 (sum_log_test_suite_test_sum_log_basic)
 PASS - test_sum_log_basic in 0.002 seconds
===================================================================
START - test_sum_log_negative
I: Logger sum: -10 + -20 = -30
W: Example warning-level message
E: Example error-level message (demonstration only)
I: Sum inputs
I: f6 ff ff ff ec ff ff ff |........

   Assertion succeeded at CMAKE_SOURCE_DIR/src/sum_test.c:16 (sum_log_test_suite_test_sum_log_negative)
 PASS - test_sum_log_negative in 0.002 seconds
===================================================================
START - test_sum_log_zero
I: Logger sum: 0 + 0 = 0
W: Example warning-level message
E: Example error-level message (demonstration only)
I: Sum inputs
I: 00 00 00 00 00 00 00 00 |........

   Assertion succeeded at CMAKE_SOURCE_DIR/src/sum_test.c:24 (sum_log_test_suite_test_sum_log_zero)
 PASS - test_sum_log_zero in 0.002 seconds
===================================================================
TESTSUITE sum_log_test_suite succeeded

------ TESTSUITE SUMMARY START ------

SUITE PASS - 100.00% [sum_log_test_suite]: pass = 3, fail = 0, skip = 0, total = 3 duration = 0.006 seconds
 - PASS - [sum_log_test_suite.test_sum_log_basic] duration = 0.002 seconds
 - PASS - [sum_log_test_suite.test_sum_log_negative] duration = 0.002 seconds
 - PASS - [sum_log_test_suite.test_sum_log_zero] duration = 0.002 seconds

------ TESTSUITE SUMMARY END ------

===================================================================
PROJECT EXECUTION SUCCESSFUL
```

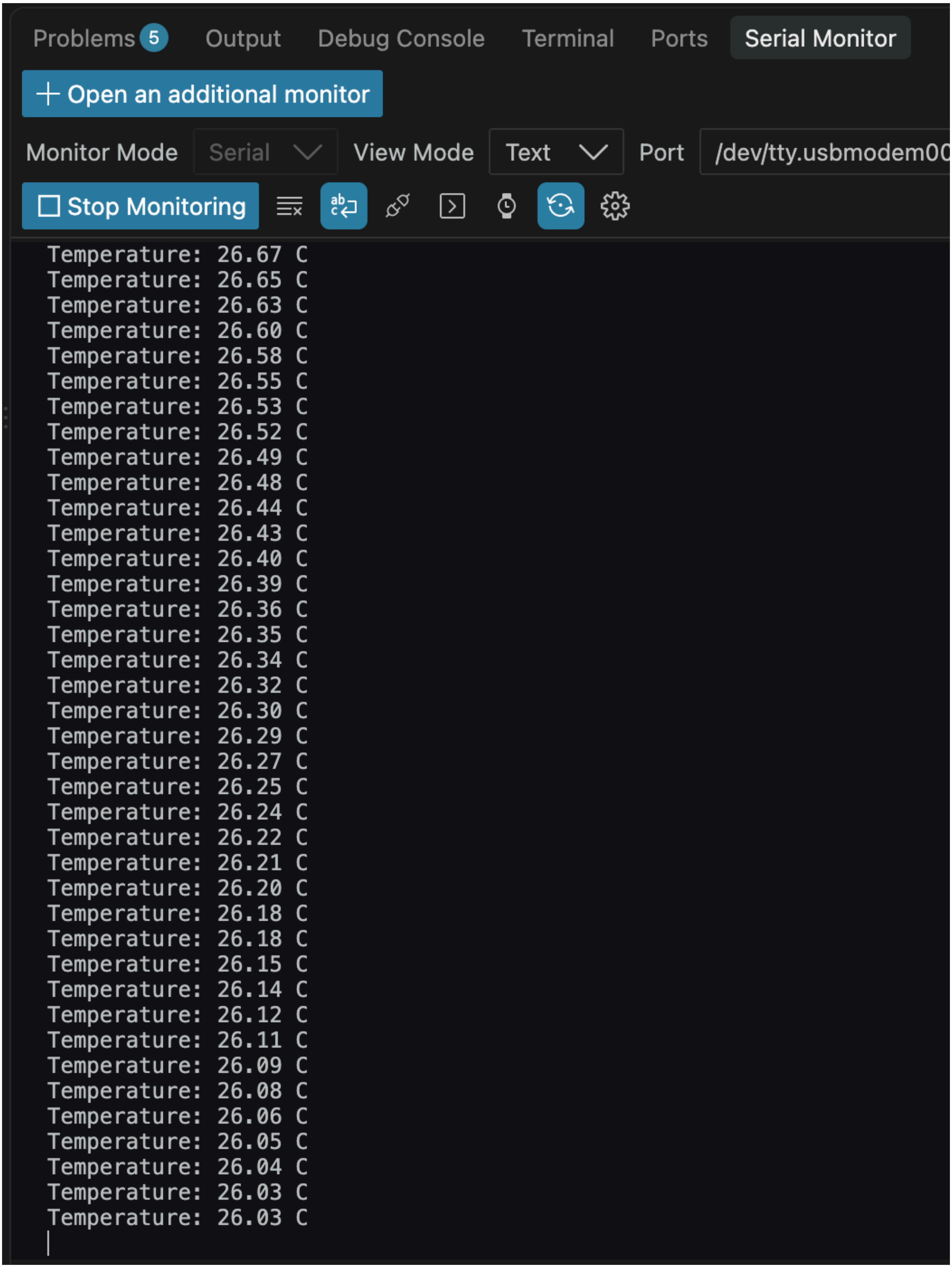
B: run in root directory (LAB_1)

```
master@Mac blinky % west twister -T tests/SUM_UNIT_TEST \
  -p qemu_cortex_m3 \
  --inline-logs -v
INFO    - Using Ninja..
INFO    - Zephyr version: ncs-v3.4.0
INFO    - Using 'zephyr/gnu' toolchain variant.
INFO    - Building initial testsuite list...
INFO    - Built testsuite list in 0.00 seconds
INFO    - Writing JSON report /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/twister-out/testplan.json
INFO    - Adding tasks to the queue...
INFO    - Added initial list of jobs to queue
INFO    - 1/1 qemu_cortex_m3/ti_lm3s6965 sum_log.test                                    PASSED (qemu 1.017s <zephyr/gnu>)

INFO    - 1 test scenarios (1 configurations) selected, 0 configurations filtered (0 by static filter, 0 at runtime).
INFO    - 1 of 1 executed test configurations passed (100.00%), 0 built (not run), 0 failed, 0 errored, with no warnings in 11.72 seconds.
INFO    - 3 of 3 executed test cases passed (100.00%) on 1 out of total 1514 platforms (0.07%).
INFO    - 1 test configurations executed on platforms, 0 test configurations were only built.
INFO    - Saving reports...
INFO    - Writing JSON report /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/twister-out/twister.json
INFO    - Writing xunit report /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/twister-out/twister.xml...
INFO    - Writing xunit report /Users/master/ESE5180/lab0-zephyr-skeleton/blinky/twister-out/twister_report.xml...
INFO    - Run completed
```

8. Adding a Peripheral (BME280)

(8.1) Print out the temperature. Take screenshots of your logging output. Commit your Zephyr application to your GitHub repository.



(8.2) Create a Ztest to test if the device tree is set up and run sanity checks. Take and embed screenshots of your Ztest output for your README.md. Commit your Zephyr application to your GitHub repository.

```
master@Mac blinky % west build -d build_bme280_test -t run
-- west build: running target run
[0/1] To exit from QEMU enter: 'CTRL+a, x'[QEMU] CPU: cortex-m3
qemu-system-arm: warning: nic stellaris_enet.0 has no peer
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE bme280_test_suite
===================================================================
START - test_devicetree_configuration

   Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:10 (bme280_test_suite_test_devicetree_configuration)

   Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:15 (bme280_test_suite_test_devicetree_configuration)

   Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:19 (bme280_test_suite_test_devicetree_configuration)
 PASS - test_devicetree_configuration in 0.002 seconds
===================================================================
START - test_temperature_datasheet_example

   Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:38 (bme280_test_suite_test_temperature_datasheet_example)
 PASS - test_temperature_datasheet_example in 0.001 seconds
===================================================================
START - test_temperature_sanity

   Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:61 (bme280_test_suite_test_temperature_sanity)

   Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:65 (bme280_test_suite_test_temperature_sanity)
 PASS - test_temperature_sanity in 0.001 seconds
===================================================================
TESTSUITE bme280_test_suite succeeded

------ TESTSUITE SUMMARY START ------

SUITE PASS - 100.00% [bme280_test_suite]: pass = 3, fail = 0, skip = 0, total = 3 duration = 0.004 seconds
 - PASS - [bme280_test_suite.test_devicetree_configuration] duration = 0.002 seconds
 - PASS - [bme280_test_suite.test_temperature_datasheet_example] duration = 0.001 seconds
 - PASS - [bme280_test_suite.test_temperature_sanity] duration = 0.001 seconds

------ TESTSUITE SUMMARY END ------

===================================================================
PROJECT EXECUTION SUCCESSFUL
```

9. Teaching Team Checkoff